The Faculty of Engineering
The Maersk Mc-Kinney Moller Institute
Report/Assignment Cover Page

Student Name:	**Student ID Number:**
Jacob F Larsen	4101586

Class and Year:	Master of Science in Engineering Drone Technology and Autonomous Systems 2025–2027
Subject Code and Name:	T550021101 Kunstig intelligens værktøjer, (F26)
Lecturer Name(s):	Smith Khare and Vinay Chakravarthi Gogineni
Title of Report/Assignment:	PAPER TITLE
Submission Date:	04-06-2026

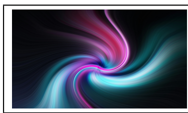
Academic Integrity and Plagiarism

Plagiarism is the act of copying, including or directly quoting from, the work of another without adequate acknowledgment. All work submitted by students for assessment purposes is accepted on the understanding that it is their own work and written in their own words except where explicitly referenced using the correct format. **For example, you must NOT copy information, ideas, portions of text, data, figures, computer programs, etc. from anywhere without giving a reference to the source.** Sources can include websites, other students' work, books, journal articles, reports, etc. Self-plagiarism or auto-plagiarism is where a student re-uses work previously submitted to another course within the SDU or other Universities.

You must ensure that you have read the SDU regulations relating to plagiarism.

I have read and understood the SDU code of practice on plagiarism and confirm that the content of this document is my own work and has not been plagiarized.

Student's signature



Contents

Abstract

Automated detection of epileptic seizures from Electroencephalography (EEG) data is critical for improving clinical workflows, patient care, and targeted neuromodulation therapies. Traditional visual inspection of massive EEG volumes by neurologists is slow, labor-intensive, and prone to human error. To address this, we investigate a hybrid deep learning framework combining a Convolutional Neural Network (CNN) and a Long Short-Term Memory (LSTM) network to automatically extract local structural patterns and track temporal dependencies in sequential time-series data.

The proposed CNN-LSTM architecture is rigorously evaluated on the Bonn University EEG dataset for binary classification between healthy states and active seizures. To assess robustness on more complex temporal structures, the model is further benchmarked against a secondary hand-gesture dataset. Minimal preprocessing, featuring bandpass filtering and z-score normalization, is applied to preserve anomalous neural spikes. For the EEG task, data augmentation is achieved via 1-second segmenting with a 75% overlap, utilizing stratified splits and weighted cross-entropy loss to combat class imbalances. Hyperparameters are optimized using a comprehensive 1152-configuration grid search followed by 5-fold cross-validation.

Experimental results demonstrate that the optimal CNN-LSTM model achieves a high classification reliability on the EEG dataset, yielding an average F1-score of 0.9895 and an accuracy of 99.30%, with an exceptionally low error rate (0.26% false negatives and 0.43% false positives). While a baseline logistic regression model achieves comparable performance on the relatively simple EEG dataset (97% accuracy), the CNN-LSTM significantly outperforms it on the complex gesture dataset, achieving an F1-score of 0.9562 and an accuracy of 95.70% compared to the baseline's 80% accuracy. These findings demonstrate the superior capability of the CNN-LSTM architecture in modeling intricate, long-running time-domain signals.

1 Introduction

Epilepsy is a common neurological disorder characterized by recurrent seizures. Traditionally, neurologists identify these seizures by visually inspecting massive volumes of Electroencephalography (EEG) data, a process that is slow, labor-intensive, and prone to human error.

Automated seizure detection using EEG data is critical for:

- Clinical Support: Assisting doctors by continuously monitoring patients and alerting hospital staff.
- Patient Care: Powering wearable devices that provide real-time warnings to reduce injury risks.
- Targeted Therapy: Enabling closed-loop neurostimulation systems that automatically trigger treatment upon detecting abnormal brain activity.

Seizure detection refers to the task of identifying patterns in EEG signals that indicate the presence of an epileptic seizure. Because machine and deep learning excel at processing complex brain patterns without manual feature engineering, this project investigates a hybrid CNN-LSTM architecture. By combining the feature extraction power of Convolutional Neural Networks (CNNs) with the temporal tracking strengths of Long Short-Term Memory (LSTM) networks, this approach aims to accurately differentiate healthy brain signals from active epileptic seizures.

2 Related Works

Early seizure detection systems relied on manually extracted statistical, spectral, or wavelet-based features paired with traditional classifiers like support vector machines, random forests, and logistic regression. While effective on smaller datasets, these approaches depended heavily on manual feature engineering and struggled to generalize across different patients.

To overcome these limitations, the current state of the art has shifted toward deep learning architectures, notably transformer-based models, graph neural networks (GNNs), and various hybrid configurations, as reviewed by Xu et al. [1] and Garg et al. [2]. Transformer-based models have gained popularity due to their self-attention mechanism, which captures long-range dependencies across the time domain. This offers a distinct advantage over recurrent neural networks (RNNs) like GRUs or LSTMs, which must process inputs sequentially and often struggle with long-term memory. To optimize this approach, recent studies have combined a transformer with varying types of convolutional networks. The works of Dong et al use a 2D-CNN [3], while Zhou et al use convolutional block attention [4].

Alternatively, GNNs have emerged as a powerful tool for analyzing multi-channel EEG readings. Because EEG channels exhibit complex spatial and functional relationships, GNNs can explicitly model these interactions through graph structures, allowing information to propagate between anatomically or functionally connected electrodes, which has been explored by Jibon et al [5].

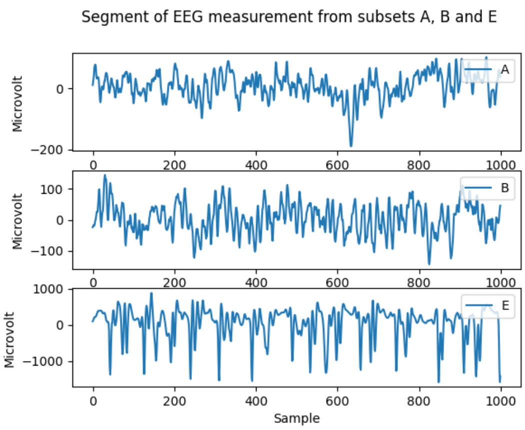
Despite their high classification performance, both transformer- and GNN-based models require massive datasets, carry heavy computational overhead, and demand prolonged training times.

Consequently, this paper adopts a hybrid CNN-LSTM model. This architecture utilizes a CNN to automatically extract spatial features from the raw input, which are then passed to an LSTM to learn sequential, temporal relationships. This design strikes a practical balance between robust feature extraction and high classification accuracy without the prohibitive computational demands of transformer or GNN architectures. Furthermore, a

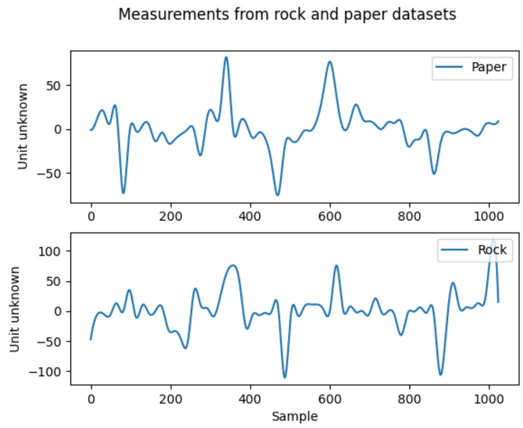
GNN approach would yield little benefit in this study, as the available dataset consists of single-channel EEG recordings that lack the inter-channel relationships GNNs are designed to exploit.

3 Methodology

3.1 Dataset



(a) Plot of one measurement from each of the subsets A, B and E. Only the first 1000 samples are shown, as there is no further structure to the data.



(b) Plot of one measurement from the paper and rock gesture dataset. The unit of the measurement is unknown, and no information could be found about the dataset.

Figure 1: Plots illusttrating the structure of the data in the two datasets used in the project.

The primary dataset used in this project is the Bonn University EEG dataset, which comprises single-channel EEG recordings from five healthy individuals and five epilepsy patients [6]. The data is divided into five subsets (A through E), each containing 100 randomized, 23.6-second segments sampled at 173.61 Hz (4,097 datapoints per measurement).

- Sets A & B: Recorded from healthy subjects with eyes open and closed, respectively.
- Sets C & D: Recorded from epilepsy patients during seizure-free intervals using two different electrode placements.
- Set E: Recorded from epilepsy patients during active seizure activity.

To focus the project on binary classification between healthy states and active seizures, only subsets A, B, and E are utilized. Consequently, the resulting dataset is unbalanced, containing twice as many healthy measurements as active seizure ones. Representative waveforms from these three subsets are displayed in fig. 1a.

3.1.1 Gesture dataset

To evaluate the model's performance on a more complex task, a secondary dataset featuring hand gestures is introduced. Each measurement consists of a single-channel, 4-second recording containing $1,024$ samples captured at a sampling frequency of 256 Hz. While the original dataset includes three gestures (rock, paper, and scissors), this study restricts analysis to the binary classification of "paper" versus "rock". Each class contains approximately 2900 samples. The underlying units and recording methodology for this dataset are unspecified. Sample measurements for both classes are visualized in fig. 1b.

3.2 Data preprocessing

Minimal preprocessing is done on the data in this project, as the intention is for the model to be trained on the time-series data. First, a bandpass filter with a passband of 0.5 Hz to 40 Hz is applied to remove artifacts introduced during data collection (only applied to EEG dataset). Next, each data point is normalized to its z-score (applied to both datasets). This step is important because the Convolutional Neural Network is very sensitive to large variations in signal amplitude; normalizing the data removes these scaling discrepancies, allowing the network to focus on underlying structural patterns. Z-score normalization is specifically chosen so that statistically anomalous spikes are preserved as distinct, high-amplitude features in the processed output.

3.3 Model architecture

The chosen model architecture for this task is a combined Convulational Neural Network and a Long Short-Term Memory (CNN-LSTM) model. The idea behind this is that the CNN model will be applied to the input data, where it will learn the most significant spatial features during training and then extract those, as the convolutional filters are passed over it. The LSTM will then learn the time-dependent relationships between those features. Finally, a fully connected layer is used to convert the LSTM output to the two target classes. The model structure and parameters can be seen in fig. 2.

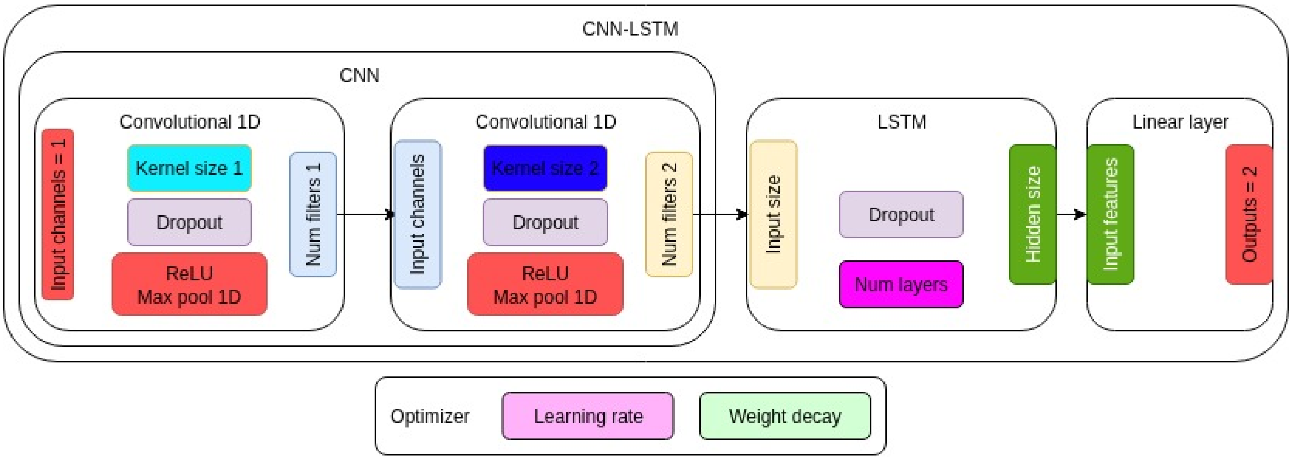


Figure 2: Structure of the implemented CNN-LSTM. All parts in red boxes are non-configurable, while boxes of all other colors contain parameters that will be varied in a grid search. The dropout parameter is kept the same in all layers, in the interest of reducing the number of configurations searched.

The CNN model is a sequential model consisting of two one-dimensional convolutional layers. These two layers each learn and extract a number of features by passing a kernel over their respective inputs. To help the CNN model learn more stably, the ReLU activation layers introduce non-linearities and prevent vanishing gradient problems, while the max pooling layers serve to only forward the most significant features while also halving the output size, such that the LSTM will be operating on input lengths one quarter the length of the input to the CNN. The dropout layers help both the CNN and the LSTM in preventing overfitting to the training data.

The LSTM is a Recurrent Neural Network (RNN), containing hidden layers comprised of memory cells, which are gated with three gates; an input, output and a forget gate. These gates serve to solve the issue of vanishing gradients, which is an issue that plagues traditional RNNs, when having to retain information for a long time. The gates control how much new information is added to the memory cell (input), how much of the old information is discarded (forget), and how much of the currently known information to output (output). In combination, they allow an LSTM model to retain relevant information over many timesteps.

To find the best combination of model hyper-parameters a grid search was performed.

3.4 Data segmentation

To reduce model input size and expand the training set, all measurements are divided into 1-second segments. This reduction does not degrade data quality because the task focuses on state detection (identifying an active seizure) rather than capturing long-term temporal structures. Shorter inputs also lower model complexity, reducing the memory retention required for the LSTM and allowing the CNN to utilize fewer filters.

To further augment the training data, these segments are generated with a 75% overlap, transforming a 2-second recording window into five distinct 1-second samples. This overlap enhances robustness, encouraging the model to recognize seizure markers based on localized characteristics rather than their spatial position within the sequence. Validation and test sets are segmented identically but without overlap to avoid artificially inflating evaluation metrics.

For gesture classification, the data is not segmented, as most of the information is in the evolution of the signal over time.

3.4.1 Data leakage and class imbalance

To avoid data leakage related to segmentation, all train/test splits are done before the signal is segmented, that way none of the validation data has a possibility of ending up in the training data, or vice-versa.

To account for class imbalances in the dataset, all train/test splits are stratified such that each split will have the same proportion of classes (±1 sample). This is also true for the 5-fold cross validation, where a stratified implementation is used. This is important because the dataset contains twice as many healthy measurements compared to seizure measurements, so if either is proportionally misrepresented in the validation data, the evalutation may show misleading results.

Stratification is also used for gesture classification, even though it is not strictly necessary as there are an approximately equal number of samples of each class.

3.5 Model training

Model training utilizes cross-entropy loss as the criterion. To account for class imbalance, the loss function is weighted, assigning double the weight to the active seizure class. Optimization is driven by the Adam optimizer, which combines adaptive learning rates with momentum to adjust individual weights and suppress gradient oscillations. To further stabilize convergence, a "Reduce-on-Plateau" learning rate scheduler dynamically lowers the learning rate when the validation loss plateaus, helping the model settle into a local minimum.

To mitigate overfitting, an early stopping mechanism monitors validation loss at each epoch, saving the model parameters only upon improvement. If the validation loss fails to improve for a predetermined number of epochs, training terminates, and the best-performing weights are restored for evaluation.

Hyperparameter tuning is conducted via a grid search evaluated through hold-out validation. While accuracy, precision, recall, and the confusion matrix are recorded for each configuration, the F1-score is used as the primary metric to determine the optimal model. The F1-score is the harmonic mean of precision and recall, and balances the penalties for false positives and false negatives, both critical in clinical seizure detection, while robustly handling the dataset's class imbalance. The equations for accuracy, precision, recall and F1-score can be seen in table 1.

Accuracy $= \frac{TP+TN}{TP+FP+TN+FN}$	Precision $= \frac{TP}{TP+FP}$	Recall $= \frac{TP}{TP+FN}$	$F1 = 2\frac{\text{precision}\cdot\text{recall}}{\text{precision}+\text{recall}}$

Table 1: Performance metric equations. TP is the number of true positives, FP is the number of false positives, TN is the number of true negatives and FN is the number of false negatives. All these numbers can be found in the confusion matrix after a model has been evaluated.

4 Results

4.1 Grid search

The grid search algorithm searched through 1152 different combinations of hyper-parameters such as kernel sizes in the CNN and the hidden size in the LSTM. The tested parameters can be seen in table 2. During grid search, each model was validated using hold-out validation, with 10% of data being used for final evaluation, 18% for validation during training and 72% for training. All train/test splits were seeded such that each model had the same data for all three scenarios. During grid search the early stopping mechanism was configured to stop the training after five epochs, if validation loss did not improve.

Table 2: Model hyper parameters that were searched using grid search. Not every combination was tested, but a subjectively selected subset. See fig. 2 for where each parameter is used.

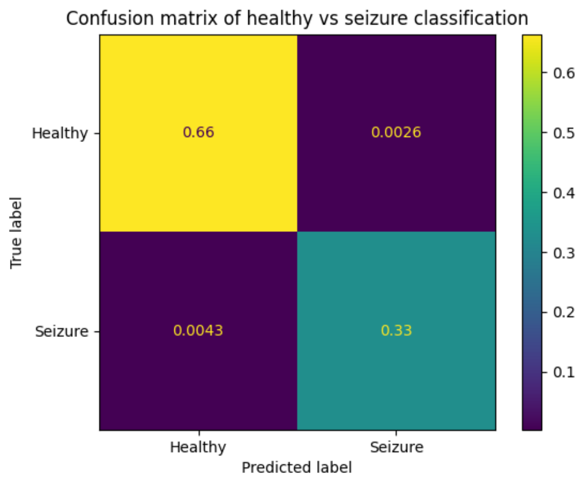
Parameter	Kernel size 1	Num filters 1	Kernel size 2	Num filters 2	Num layers	Hidden size	Dropout	Learning rate	Weight decay
Values	5, 7, 9, 11, 13	32, 64, 128	5, 7, 9, 11, 13	64, 128, 256	1, 2, 3	64, 128, 256	0.2, 0.5, 0.6	1e-3, 1e-4	1e-3, 1e-4, 1e-5

4.2 Cross validation

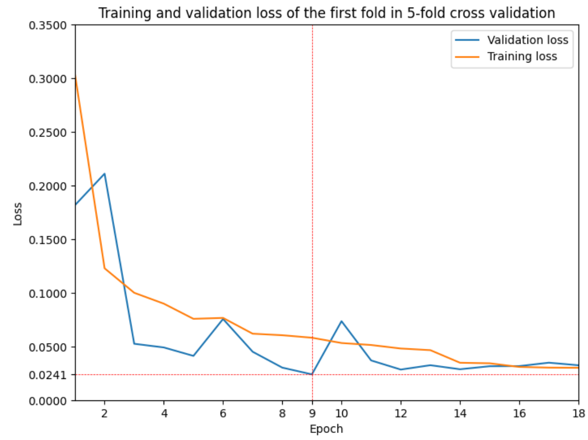
The five top-performing models from the grid search (ranked by F1-score) were evaluated using five-fold cross-validation. To allow sufficient opportunity for optimization, the early stopping patience was set to nine epochs.

The parameters and cross-validation metrics (averaged across all five folds) for the optimal seizure detection model are presented in table 3. fig. 3a displays the averaged, normalized confusion matrix for this model. The model achieved a low error rate of 0.69%, consisting of 0.26% false negatives and 0.43% false positives.

The training and validation loss curves for the first fold are illustrated in fig. 3b. The dashed red line indicates the epoch with the lowest validation loss; the early stopping mechanism saved the weights from this epoch and restored them post-training for the final evaluation of the fold.



(a) Confusion matrix of the seizure detection model, averaged over all 5 folds.



(b) Training and validation loss during training of the first fold in 5-fold cross validation. The red lines show the exact epoch where the model obtained the best validation loss, as well as the exact value of the loss.

Figure 3: Confusion matrix and training & validation loss of the best seizure detection model.

Table 4: Seizure classification results of logistic regression model, implemented by group member

(a) Performance metrics from logistic regression model.

Precision	Recall	F1-score	Accuracy	Support
0.97	0.96	0.97	0.97	60

(b) Confusion matrix from logistic regression model.

	PP	PN
AP	32	0
AN	2	26

Table 3: Performance metrics and parameters of the best seizure detection model found by five-fold cross validation.

Kernel size 1	Num filters 1	Kernel size 2	Num filters 2	Num layers	Hidden size	Dropout	Learning rate	Weight decay
5	64	9	64	2	64	0.5	1e-3	1e-4

F1-score	Precision	Recall	Accuracy
0.9895	0.9921	0.9870	0.9930

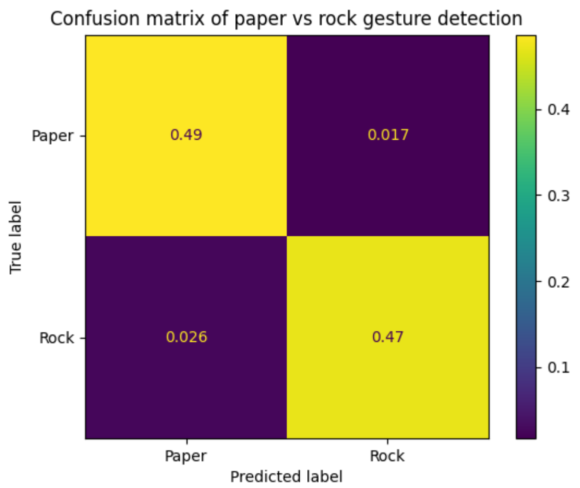
4.3 Comparison with logistic regression

While this model behaves very nicely on the Bonn University EEG dataset, this is also quite a simple dataset, with very clear separations between the classified labels, meaning that a much simpler model would have been able to reach comparable results. A group member trained a logistic regression model, using only statistical features of the dataset, their results can be seen in table 4. As seen in the table, their model reaches a result remarkably close to the CNN-LSTM model despite being much simpler. This can mostly be explained by the simplicity of the dataset.

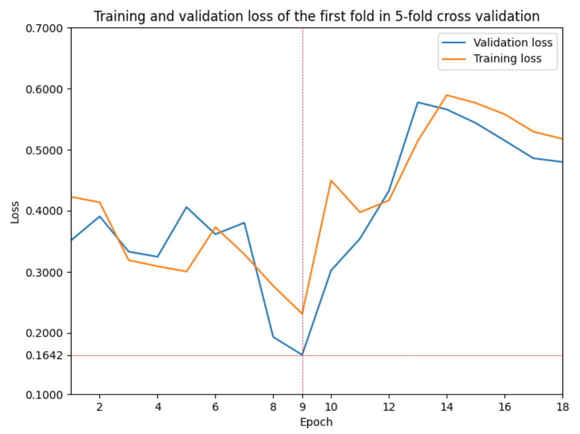
Table 5: Performance metrics and parameters of the best model for gesture classification found by five-fold cross validation

Kernel size 1	Num filters 1	Kernel size 2	Num filters 2	Num layers	Hidden size	Dropout	Learning rate	Weight decay
11	128	7	256	3	128	0.5	1e-3	1e-4

F1-score	Precision	Recall	Accuracy
0.9562	0.9652	0.9478	0.9570



(a) Confusion matrix of the gesture classification model, averaged over all 5 folds.



(b) Training and validation loss during training of the first fold in 5-fold cross validation. The red lines show the exact epoch where the model obtained the best validation loss, as well as the exact value of the loss.

Figure 4: Confusion matrix and training & validation loss of the best performing gesture classification model.

4.4 Comparing gesture classification capabilities

To test how the two models compare on a more advanced dataset, both models were tasked with binary classification on a gesture dataset. With the gesture dataset, the same five model configurations as before were validated with five-fold cross validation. This is justified by the relatively large variation in hyper-parameter configurations of those five models.

This dataset seems to favor a more complex model, with larger kernel sizes and layer sizes. This also makes sense in the context that this data is much more complex and spans a longer duration, so the CNN can find more significant features and the LSTM is required to retain information for much longer. Gesture classification performance metrics of the CNN-LSTM model can be seen in table 5 along with the hyper-parameters of the model. Confusion matrix and loss curves can be seen in fig. 4.

As can be seen by comparing the logistic regression model (table 6) and the CNN-LSTM model (table 5 and fig. 4a), the more advanced CNN-LSTM model is better suited to this type of task, where long-running time-domain signals need classifying, but that also comes at the cost of higher complexity and larger computational costs.

Table 6: Gesture classification results of logistic regression model, implemented by group member

(a) Performance metrics from logistic regression model doing gesture classification.

Precision	Recall	F1-score	Accuracy	Support
0.81	0.80	0.80	0.80	1162

(b) Confusion matrix from logistic regression model.

	PP	PN
AP	429	168
AN	68	497

4.5 Comparing to state of the art

Comparing the results of the seizure detection CNN-LSTM model with those of the models reviewed in a survey by Garg et al [2], shows that this model reaches comparable performance, though the paper also shows that others have used the same structure to reach even better results, with accuracy of up to 100%, as in the works of Mallick and Baths [7].

5 Discussion and Conclusions

The CNN-LSTM architecture demonstrates strong performance in the classification of time-domain signals. For seizure detection, the model achieved an average F1-score of 0.989 and an accuracy of 99.3%, indicating a high degree of classification reliability. In the gesture detection task, the CNN-LSTM model attained an average F1-score of 0.956 and an accuracy of 96%, which also reflects strong performance. However, the hyper-parameter configurations explored in this project suggest that further performance improvements may be attainable. In particular, gesture detection appears to benefit from increased model complexity, including larger convolutional kernels, a greater number of layers, and additional filters.

A notable limitation of CNN-LSTM architectures is their computational cost, both during training and inference. This cost increases substantially as network depth and layer size grow, potentially limiting the feasibility of deploying such models in real-time applications. This concern is particularly relevant for seizure detection, where low-latency processing may be critical. Nevertheless, the impact of this limitation can be mitigated through the use of appropriate hardware accelerators, such as graphics processing units (GPUs), which can significantly reduce both training and inference times.

6 Reflection

In this project a CNN-LSTM model was implemented to detect seizures in the Bonn University EEG dataset. A grid search algorithm was executed to find the best combination of hyper-parameters. Care was taken to avoid data leakage, and the class imbalance of the dataset was also considered. Minimal preprocessing was done to the data, but it was segmented, such that the model was trained on one-second chunks instead of whole measurements, this increased the amount of training data, and lowered the complexity of the input data, enabling simpler models.

The best 5 configurations found by the grid search algorithm were then evaluated with 5-fold cross validation, and the best model of those achieved an average F1-score of 0.9895 and accuracy of 99.3%. The model was compared to a simpler logistic regression model, which reached comparable results, due to the simplicity of the dataset used.

The same 5 best configurations for EEG classification were also evaluated in gesture classification, where the 5-fold cross validation revealed a different configuration to be best. In this task, the best model obtained an average F1-score of 0.9562 and an accuracy of 95.7%. Comparing these results to those of the logistic regression model shows the more complex CNN-LSTM to be better suited for gesture detection. This is likely due to the more complex data, where statistical features can not capture all of the information, but the CNN-LSTM is better able to find relevant features to extract during training, and learn the relationships between them.

References

[1] J. Xu et al., "Eeg-based epileptic seizure detection using deep learning techniques: A survey," *Neurocomputing*, vol. 610, p. 128 644, 2024, ISSN: 0925-2312. DOI: `https://doi.org/10.1016/j.neucom.2024.128644` [Online]. Available: `https://www.sciencedirect.com/science/article/pii/S0925231224014152`

[2] M. Garg, J. Kaur, and N. R. Prakash, "Advanced machine learning and deep learning approaches for accurate epileptic seizure detection using eeg signals," *International Journal of Imaging Systems and Technology*, vol. 36, no. 3, e70342, 2026, e70342 IMA-25-1829. DOI: `https://doi-org.proxy1-bib.sdu.dk/10.1002/ima.70342` eprint: `https://onlinelibrary-wiley-com.proxy1-bib.sdu.dk/doi/pdf/10.1002/ima.70342`. [Online]. Available: `https://onlinelibrary-wiley-com.proxy1-bib.sdu.dk/doi/abs/10.1002/ima.70342`

[3] C. Dong, D. Sun, Z. Zhang, and B. Luo, "Eeg-based patient-specific seizure prediction based on spatial–temporal hypergraph attention transformer," *Biomedical Signal Processing and Control*, vol. 100, p. 107 075, 2025, ISSN: 1746-8094. DOI: `https://doi.org/10.1016/j.bspc.2024.107075` [Online]. Available: `https://www.sciencedirect.com/science/article/pii/S1746809424011339`

[4] S. Zhou, Y. Li, J. Ma, and W. Zhou, "Efficient seizure detection using swin transformer integrated with convolutional block attention," *Neural Computing and Applications*, vol. 38, no. 10, p. 365, May 2026, ISSN: 1433-3058. DOI: `10.1007/s00521-026-12025-y` [Online]. Available: `https://doi.org/10.1007/s00521-026-12025-y`

[5] F. A. Jibon et al., "Sequential graph convolutional network and deeprnn based hybrid framework for epileptic seizure detection from eeg signal," *Digital Health*, vol. 10, p. 20 552 076 241 249 874, 2024.

[6] W. Chen et al., "An automated detection of epileptic seizures eeg using cnn classifier based on feature fusion with high accuracy.," *BMC medical informatics and decision making*, vol. 23, no. 1, p. 96, 2023. DOI: `10.1186/s12911-023-02180-w`

[7] S. Mallick and V. Baths, "Novel deep learning framework for detection of epileptic seizures using eeg signals," *Frontiers in Computational Neuroscience*, vol. Volume 18 - 2024, 2024, ISSN: 1662-5188. DOI: `10.3389/fncom.2024.1340251` [Online]. Available: `https://www.frontiersin.org/journals/computational-neuroscience/articles/10.3389/fncom.2024.1340251`